

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

Interpretation. $D = 1$: the patient has the disease. $T = 1$: the test result is positive.

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

Idea. The disease causes the test result: the causal graph is $D \rightarrow T$. The problem gives the probabilities along the arrow: $p(D)$ and $p(T \mid D)$.

Rules. Let $A \rightarrow B$ be an arrow of the causal graph: B depends directly on A . Each rule gives a formula for a probability that is not given:

R1. $p(A \mid B)$, against the arrow: $p(A \mid B) = \frac{p(A, B)}{p(B)},$

R2. $p(A, B)$, along the arrow: $p(A, B) = p(A) p(B \mid A),$

R3. $p(B)$, the parent A is not fixed: $p(B) = \sum_a p(A = a, B) \quad (\text{for continuous } A: \int da),$

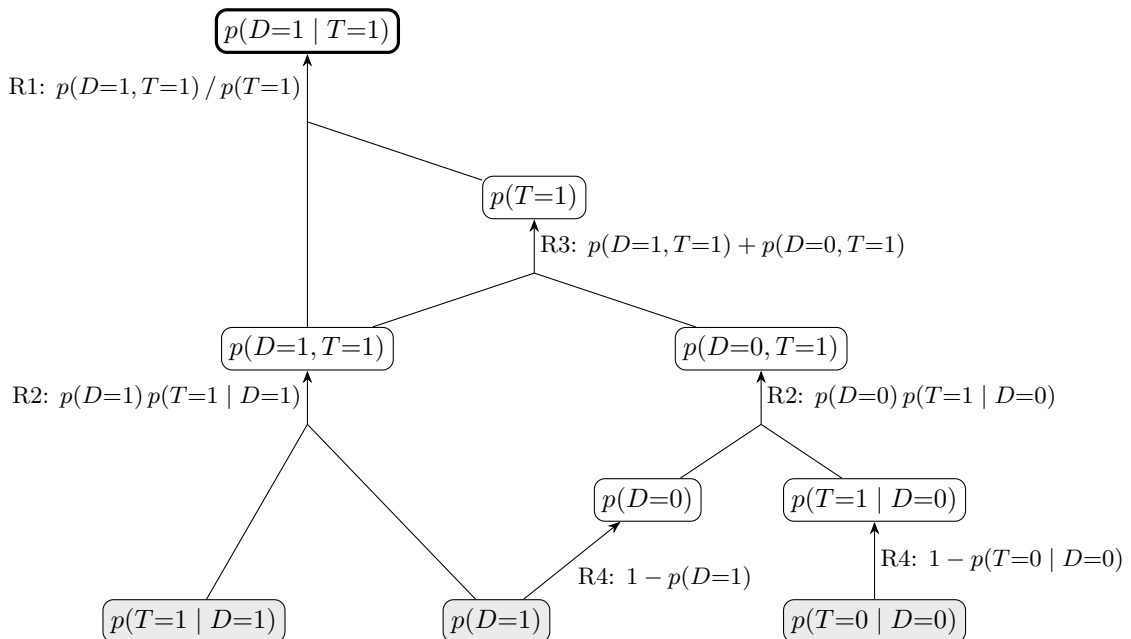
R4. The other value of a binary variable: $p(A = 0 \mid C) = 1 - p(A = 1 \mid C).$

R1, R2, and R3 together give Bayes' theorem:

$$p(A \mid B) = \frac{p(A) p(B \mid A)}{\sum_a p(A = a) p(B \mid A = a)}.$$

Computation graph. Put the goal at the top. For each node that is not given, apply the rule for its form. Put the terms of the formula below the node as new nodes. Stop when all nodes at the bottom are given. Then calculate the nodes from the bottom to the top.

Solution. *Computation graph.* Each arrow shows the rule and the formula for the node above it. Shaded nodes are given.



Step 1. Find $p(D = 0)$ with R4.

$$p(D = 0) = 1 - p(D = 1) = 1 - 0.0001 = 0.9999.$$

Step 2. Find $p(T = 1 \mid D = 0)$ with R4.

$$p(T = 1 \mid D = 0) = 1 - p(T = 0 \mid D = 0) = 1 - 0.99 = 0.01.$$

Step 3. Find $p(D = 1, T = 1)$ with R2.

$$p(D = 1, T = 1) = p(D = 1) p(T = 1 \mid D = 1) = 0.0001 \cdot 0.99 = 0.000099.$$

Step 4. Find $p(D = 0, T = 1)$ with R2.

$$p(D = 0, T = 1) = p(D = 0) p(T = 1 \mid D = 0) = 0.9999 \cdot 0.01 = 0.009999.$$

Step 5. Find $p(T = 1)$ with R3.

$$\begin{aligned} p(T = 1) &= \sum_D p(D, T = 1) = p(D = 1, T = 1) + p(D = 0, T = 1) \\ &= 0.000099 + 0.009999 = 0.010098. \end{aligned}$$

Step 6. Find $p(D = 1 \mid T = 1)$ with R1.

$$\begin{aligned} p(D = 1 \mid T = 1) &= \frac{p(D = 1, T = 1)}{p(T = 1)} = \frac{0.000099}{0.010098} = \frac{99}{10098} \quad (\text{multiply by } 10^6) \\ &= \frac{99}{99 \cdot 102} = \frac{1}{102} \approx 0.0098. \end{aligned}$$

Conclusion. The probability increases from 0.01% to only 1%. In 10 000 people, the test finds $10\,000 \cdot 0.000099 \approx 1$ sick person and $10\,000 \cdot 0.009999 \approx 100$ healthy people. Thus most positive results of a mass test for a rare disease are false. A second positive independent test uses $p(D = 1 \mid T_1 = 1) = 1/102$ as the prior:

$$\begin{aligned} p(D = 1 \mid T_1 = 1, T_2 = 1) &= \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \left(1 - \frac{1}{102}\right) \cdot 0.01} \\ &= \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \frac{101}{102} \cdot 0.01} \\ &= \frac{0.99}{0.99 + 101 \cdot 0.01} \quad (\text{multiply by } 102) \\ &= \frac{0.99}{0.99 + 1.01} = \frac{0.99}{2} = 0.495. \end{aligned}$$

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D) \mid P \quad D$	$p(H = 1 \mid P) \mid P$	$p(T = 1 \mid A) \mid A$
0.999	0.9	0.95
0.9	0.2	0.5
0.9		
0.01		

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

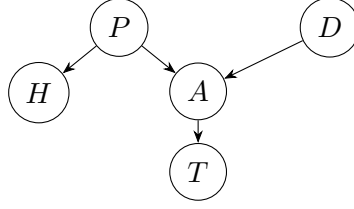


Figure 1: Causal relations for Problem 2

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

Idea. Use the method of Problem 1 on a larger graph.

Joint distribution. Apply the chain rule one variable at a time, from the children to the parents. A missing arrow means no direct influence: given its parents, a node does not depend on the other nodes in the condition.

$$\begin{aligned}
 p(P, D, A, H, T) &= p(P, D, A, H) p(T \mid P, D, A, H) \\
 &= p(P, D, A, H) p(T \mid A) && \text{(parent of } T: A) \\
 p(P, D, A, H) &= p(P, D, A) p(H \mid P, D, A) \\
 &= p(P, D, A) p(H \mid P) && \text{(parent of } H: P) \\
 p(P, D, A) &= p(P, D) p(A \mid P, D) && \text{(parents of } A: P, D) \\
 p(P, D) &= p(P) p(D \mid P) \\
 &= p(P) p(D) && (D \text{ has no parents})
 \end{aligned}$$

Thus

$$p(P, D, A, H, T) = p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A).$$

Each factor is a given table.

Queries. Move along the edges with sums, and against the edges with Bayes' theorem:

$$p(T \mid P) = \sum_A p(T \mid A) p(A \mid P), \quad p(P \mid T) = \frac{p(P) p(T \mid P)}{\sum_{P'} p(P') p(T \mid P')}.$$

A sum over a node without children removes its factor, because $\sum_x p(x \mid \text{parents}) = 1$.

Solution.

Step 1. Find $p(P = 1 \mid H = 1)$.

$$\begin{aligned}
 p(P = 1 \mid H = 1) &= \frac{p(P = 1, H = 1)}{p(H = 1)} = \frac{p(P = 1, H = 1)}{\sum_P p(P, H = 1)} \\
 &= \frac{p(P = 1, H = 1)}{p(P = 1, H = 1) + p(P = 0, H = 1)}.
 \end{aligned}$$

Step 1a. Find $p(P, H)$ for Step 1. Sum out T, A, D :

$$\begin{aligned}
 p(P, H) &= \sum_D \sum_A \sum_T p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A) \\
 &= p(P) p(H \mid P) \sum_D p(D) \sum_A p(A \mid P, D) \sum_T p(T \mid A) && \text{(move factors out of the sums)} \\
 &= p(P) p(H \mid P) \sum_D p(D) \sum_A p(A \mid P, D) && (\sum_T p(T \mid A) = 1) \\
 &= p(P) p(H \mid P) \sum_D p(D) && (\sum_A p(A \mid P, D) = 1) \\
 &= p(P) p(H \mid P) && (\sum_D p(D) = 1).
 \end{aligned}$$

Substitute the values:

$$\begin{aligned} p(P = 1, H = 1) &= p(P = 1) p(H = 1 \mid P = 1) = 0.2 \cdot 0.9 = 0.18, \\ p(P = 0, H = 1) &= (1 - p(P = 1)) p(H = 1 \mid P = 0) \\ &= (1 - 0.2) \cdot 0.2 = 0.8 \cdot 0.2 = 0.16. \end{aligned}$$

Step 1b. Substitute Step 1a into Step 1.

$$p(P = 1 \mid H = 1) = \frac{0.18}{0.18 + 0.16} = \frac{0.18}{0.34} = \frac{9}{17} \approx 0.529.$$

Step 2. Find $p(P = 1 \mid T = 1)$.

$$\begin{aligned} p(P = 1 \mid T = 1) &= \frac{p(P = 1, T = 1)}{p(T = 1)} = \frac{p(P = 1, T = 1)}{p(P = 1, T = 1) + p(P = 0, T = 1)}, \\ p(P, T) &= p(P) p(T \mid P). \end{aligned}$$

Step 2a. Find $p(T \mid P)$ for Step 2.

$$\begin{aligned} p(T \mid P) &= \sum_A p(T, A \mid P) && \text{(sum over } A) \\ &= \sum_A p(T \mid A, P) p(A \mid P) && \text{(product rule)} \\ &= \sum_A p(T \mid A) p(A \mid P) && \text{(parent of } T: A) \\ &= p(T \mid A = 1) p(A = 1 \mid P) + p(T \mid A = 0) p(A = 0 \mid P). \end{aligned}$$

Step 2b. Find $p(A \mid P)$ for Step 2a.

$$\begin{aligned} p(A \mid P) &= \sum_D p(A, D \mid P) && \text{(sum over } D) \\ &= \sum_D p(A \mid P, D) p(D \mid P) && \text{(product rule)} \\ &= \sum_D p(A \mid P, D) p(D) && (D \text{ has no parents)} \\ &= p(A \mid P, D = 1) p(D = 1) + p(A \mid P, D = 0) p(D = 0). \end{aligned}$$

Substitute the values, with $p(D = 0) = 1 - 0.4 = 0.6$:

$$\begin{aligned} p(A = 1 \mid P = 1) &= 0.999 \cdot 0.4 + 0.9 \cdot 0.6 = 0.3996 + 0.54 = 0.9396, \\ p(A = 1 \mid P = 0) &= 0.9 \cdot 0.4 + 0.01 \cdot 0.6 = 0.36 + 0.006 = 0.366. \end{aligned}$$

Step 2c. Substitute Step 2b into Step 2a. With $p(A = 0 \mid P) = 1 - p(A = 1 \mid P)$:

$$\begin{aligned} p(T = 1 \mid P = 1) &= 0.95 \cdot 0.9396 + 0.5 \cdot (1 - 0.9396) = 0.95 \cdot 0.9396 + 0.5 \cdot 0.0604 \\ &= 0.89262 + 0.0302 = 0.92282, \\ p(T = 1 \mid P = 0) &= 0.95 \cdot 0.366 + 0.5 \cdot (1 - 0.366) = 0.95 \cdot 0.366 + 0.5 \cdot 0.634 \\ &= 0.3477 + 0.317 = 0.6647. \end{aligned}$$

Step 2d. Substitute Step 2c into Step 2.

$$\begin{aligned} p(P = 1, T = 1) &= p(P = 1) p(T = 1 \mid P = 1) = 0.2 \cdot 0.92282 = 0.184564, \\ p(P = 0, T = 1) &= p(P = 0) p(T = 1 \mid P = 0) = 0.8 \cdot 0.6647 = 0.53176, \end{aligned}$$

$$p(P = 1 \mid T = 1) = \frac{0.184564}{0.184564 + 0.53176} = \frac{0.184564}{0.716324} \approx 0.258.$$

Step 3. Find $p(P = 1 \mid T = 1, H = 1)$.

$$\begin{aligned} p(P = 1 \mid T = 1, H = 1) &= \frac{p(P = 1, H = 1, T = 1)}{p(H = 1, T = 1)} \\ &= \frac{p(P = 1, H = 1, T = 1)}{p(P = 1, H = 1, T = 1) + p(P = 0, H = 1, T = 1)}. \end{aligned}$$

Step 3a. Find $p(P, H, T)$ for Step 3. Sum out A and D :

$$\begin{aligned} p(P, H, T) &= \sum_A \sum_D p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A) \\ &= p(P) p(H \mid P) \sum_A p(T \mid A) \sum_D p(D) p(A \mid P, D) \quad (\text{move factors out of the sums}) \\ &= p(P) p(H \mid P) \sum_A p(T \mid A) p(A \mid P) \quad (\text{Step 2b}) \\ &= p(P) p(H \mid P) p(T \mid P) \quad (\text{Step 2a}). \end{aligned}$$

Substitute the values from Step 2c:

$$\begin{aligned} p(P = 1, H = 1, T = 1) &= 0.2 \cdot 0.9 \cdot 0.92282 = 0.18 \cdot 0.92282 = 0.166108, \\ p(P = 0, H = 1, T = 1) &= 0.8 \cdot 0.2 \cdot 0.6647 = 0.16 \cdot 0.6647 = 0.106352. \end{aligned}$$

Step 3b. Substitute Step 3a into Step 3.

$$p(P = 1 \mid T = 1, H = 1) = \frac{0.166108}{0.166108 + 0.106352} = \frac{0.166108}{0.27246} \approx 0.610.$$

Conclusion.

$$\underbrace{0.2}_{\text{prior}} \rightarrow \underbrace{0.529}_{H=1}, \underbrace{0.258}_{T=1}, \underbrace{0.610}_{H=1, T=1}.$$

An unhappy teacher is weak evidence: $p(T = 1 \mid P = 0) = 0.6647$ is also high. Two observations together give the strongest evidence.

3. Let $x_1 \sim \text{Poiss}(\lambda_1)$ and $x_2 \sim \text{Poiss}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$.

Idea. For $y = f(x_1, \dots, x_n)$ with discrete x_i , split the event $\{y = k\}$ into disjoint events and sum:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n) = \sum_{f(j_1, \dots, j_n) = k} \prod_i p(x_i = j_i) \quad (\text{if independent}).$$

Then simplify the sum and compare it with known distributions.

Solution. Goal: show that $p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)}(\lambda_1 + \lambda_2)^k / k!$ for all $k \geq 0$.

$$\begin{aligned}
p(x_1 + x_2 = k) &= p\left(\bigcup_{j=0}^k \{x_1 = j, x_2 = k - j\}\right) && (x_1, x_2 \geq 0) \\
&= \sum_{j=0}^k p(x_1 = j, x_2 = k - j) && (\text{disjoint events}) \\
&= \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j) && (\text{independence}) \\
&= \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} \cdot e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!} && (\text{Poisson probabilities}) \\
&= e^{-\lambda_1} e^{-\lambda_2} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && (\text{factors without } j \text{ out of the sum}) \\
&= e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && (e^x e^y = e^{x+y}) \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \frac{k!}{j! (k-j)!} \lambda_1^j \lambda_2^{k-j} && (\text{multiply and divide by } k!) \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} && (\binom{k}{j} = \frac{k!}{j! (k-j)!}) \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k && (\text{binomial theorem}).
\end{aligned}$$

This is $\text{Poiss}(\lambda_1 + \lambda_2)$.

Conclusion. A sum of independent Poisson variables is a Poisson variable, and the parameters add. By induction, this is true for any finite number of variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. Find the maximum likelihood estimate of λ .

Idea.

$$L(\lambda) = \prod_i p(x_i | \lambda) \rightarrow \ell(\lambda) = \log L(\lambda) = \sum_i \log p(x_i | \lambda) \rightarrow \ell'(\lambda) = 0 \rightarrow \ell''(\lambda) < 0.$$

The logarithm is increasing, so L and ℓ have the same maximum point.

Solution. Let $S = \sum_{i=1}^N x_i$ and $\bar{x} = S/N$.

Step 1. Write the goal.

$$\hat{\lambda}_{\text{ML}} = \arg \max_{\lambda} L(\lambda) = \arg \max_{\lambda} \ell(\lambda) \quad (\log \text{ is increasing}).$$

Step 2. Find $\ell(\lambda)$ for Step 1.

$$\begin{aligned}
 \ell(\lambda) &= \log \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \\
 &= \sum_{i=1}^N \log \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) && (\text{log of a product}) \\
 &= \sum_{i=1}^N (-\lambda + x_i \log \lambda - \log x_i!) && (\text{log of each factor}) \\
 &= -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i! && (\text{sum each term}).
 \end{aligned}$$

Step 3. Find the stationary point of ℓ .

$$\begin{aligned}
 \ell'(\lambda) &= -N + \frac{S}{\lambda}, \\
 -N + \frac{S}{\lambda} = 0 &\implies \frac{S}{\lambda} = N \implies \lambda = \frac{S}{N} = \bar{x}.
 \end{aligned}$$

Step 4. Make sure that it is a maximum.

$$\ell''(\lambda) = -\frac{S}{\lambda^2} < 0 \quad \text{if } S > 0.$$

Thus ℓ is strictly concave, and $\bar{x}$ is the global maximum. If $S = 0$, then $\ell(\lambda) = -N\lambda$ decreases, and the maximum is at $\lambda = 0 = \bar{x}$.

$$\hat{\lambda}_{\text{ML}} = \bar{x}.$$

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda \mid x_1, \dots, x_N)$.

Idea. The numerator and the denominator of Bayes' theorem contain the same function:

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

Do not calculate the integral directly:

$$\begin{aligned}
 p(X \mid \lambda) p(\lambda) &= C \cdot g(\lambda) && (C \text{ does not depend on } \lambda) \\
 g(\lambda) &= Z q(\lambda) && (q \text{ is a known density}) \\
 \int CZ q(\lambda) d\lambda &= CZ && (\text{a density integrates to 1}) \\
 p(\lambda \mid X) &= \frac{CZ q(\lambda)}{CZ} = q(\lambda).
 \end{aligned}$$

The same method gives the posterior mean.

Solution. Let $S = \sum_{i=1}^N x_i$, $a' = a + S$, and $b' = b + N$.

Step 1. Write the goal.

$$\hat{\lambda}_{\text{B}} = \mathbb{E}[\lambda \mid X] = \int \lambda p(\lambda \mid X) d\lambda.$$

Step 2. Find $p(\lambda \mid X)$ for Step 1.

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}.$$

Step 2a. Separate the constant in the numerator of Step 2.

$$\begin{aligned} p(X \mid \lambda) p(\lambda) &= \prod_{i=1}^N \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} \\ &= \frac{\prod_i e^{-\lambda} \cdot \prod_i \lambda^{x_i}}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && \text{(split the product)} \\ &= \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\prod_i \lambda^{x_i} = \lambda^{\sum_i x_i}) \\ &= \frac{b^a}{\Gamma(a) \prod_i x_i!} \cdot \lambda^S \lambda^{a-1} \cdot e^{-N\lambda} e^{-b\lambda} && \text{(group the factors)} \\ &= \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a'-1} e^{-b'\lambda}}_{g(\lambda)} && \text{(add the exponents).} \end{aligned}$$

Step 2b. Find the known density for g from Step 2a.

$$\mathcal{G}(\lambda \mid a', b') = \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} = \frac{b'^{a'}}{\Gamma(a')} g(\lambda) \implies g(\lambda) = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b').$$

Step 2c. Find the denominator of Step 2.

$$\begin{aligned} \int C g(\lambda) d\lambda &= \int C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b') d\lambda && \text{(Step 2b)} \\ &= C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda \mid a', b') d\lambda && \text{(constants out of the integral)} \\ &= C \frac{\Gamma(a')}{b'^{a'}} && \text{(a density integrates to 1).} \end{aligned}$$

Step 2d. Substitute Steps 2a–2c into Step 2.

$$\begin{aligned} p(\lambda \mid X) &= \frac{C g(\lambda)}{C \frac{\Gamma(a')}{b'^{a'}}} && \text{(Steps 2a, 2c)} \\ &= \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b')}{C \frac{\Gamma(a')}{b'^{a'}}} && \text{(Step 2b)} \\ &= \mathcal{G}(\lambda \mid a', b') = \mathcal{G}(\lambda \mid a + S, b + N). \end{aligned}$$

Step 3. Substitute Step 2d into Step 1.

$$\mathbb{E}[\lambda \mid X] = \int \lambda \mathcal{G}(\lambda \mid a', b') d\lambda = \int \lambda \cdot \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} d\lambda = \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda.$$

Step 3a. Find the integral for Step 3 with the same method. Use Step 2b with $a' + 1$ instead of a' :

$$\begin{aligned} \int \lambda^{a'} e^{-b'\lambda} d\lambda &= \int \lambda^{(a'+1)-1} e^{-b'\lambda} d\lambda \\ &= \int \frac{\Gamma(a' + 1)}{b'^{a'+1}} \mathcal{G}(\lambda \mid a' + 1, b') d\lambda && \text{(Step 2b)} \\ &= \frac{\Gamma(a' + 1)}{b'^{a'+1}} \int \mathcal{G}(\lambda \mid a' + 1, b') d\lambda && \text{(constants out of the integral)} \\ &= \frac{\Gamma(a' + 1)}{b'^{a'+1}} && \text{(a density integrates to 1).} \end{aligned}$$

Step 3b. Substitute Step 3a into Step 3.

$$\begin{aligned}
\mathbb{E}[\lambda \mid X] &= \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} \\
&= \frac{\Gamma(a' + 1)}{\Gamma(a') b'} && (b'^{a'}/b'^{a'+1} = 1/b') \\
&= \frac{a' \Gamma(a')}{\Gamma(a') b'} && (\Gamma(a' + 1) = a' \Gamma(a')) \\
&= \frac{a'}{b'} = \frac{a + S}{b + N}.
\end{aligned}$$

$$\hat{\lambda}_B = \frac{a + S}{b + N}.$$

Conclusion.

- *Conjugate prior:* gamma prior $\times$ Poisson likelihood $\rightarrow$ gamma posterior. The update only changes the parameters: $a \rightarrow a + S$ (events), $b \rightarrow b + N$ (observations).
- The estimate is a weighted mean of the prior mean and the estimate $\bar{x}$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{a}{b + N} + \frac{S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \bar{x} \xrightarrow{N \rightarrow \infty} \bar{x}.$$

- If $a, b \rightarrow 0$, then $p(\lambda) \propto \lambda^{a-1} e^{-b\lambda} \rightarrow 1/\lambda$, the non-informative prior from the notes below, and $\hat{\lambda}_B \rightarrow S/N = \bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).